



Introduction to Linear Models

From observations to predictions

PRESENTER

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Course Overview

- We will explore linear relationships in empirical datasets.
- Learn how to fit lines and evaluate prediction accuracy.



SECTION

The central question

How can a line help us make useful predictions?

Start with a familiar pattern

- Study hours and exam scores often move together.
- We want a model that describes this relationship clearly.
- The model should also help estimate an unseen score.

Learning objective

By the end of class, you will interpret the slope and intercept of a simple linear model.

A model is a useful simplification

Linear model

$$y = \beta_0 + \beta_1 x$$

The intercept β_0 is the predicted outcome at $x = 0$. The slope β_1 describes how the prediction changes as x increases by one unit.

CORE TAKEAWAY

All models are wrong, but some are useful.